Firestore Desktop Manager

Project Development Plan & Technical Specification

Project Overview

The objective is to develop a robust cross-platform desktop application using **Electron** for managing Firestore databases. The application focuses on high-efficiency data manipulation through a **Visual Query Constructor** (JQL-inspired) and **AI-assisted natural language processing** to streamline complex CRUD operations.

Core Technical Architecture

The application will utilize a dual-layer architecture to ensure security and performance:

- **Main Process (Node.js):** Handles heavy lifting via `firebase-admin` SDK, local file encryption for credentials, and AI API orchestration.
- **Renderer Process (React/Vite):** A reactive UI managing the state of the Query Constructor, data grids, and real-time validation.
- **Security:** All Service Account keys are stored using system-level keychains (`node-keytar`) or AES-256 local encryption.

Key Feature Specifications

1. Visual Query Constructor (JQL-Style)

A row-based interface allowing users to build complex queries without syntax knowledge.

- **Dynamic Dropdowns:** Fields are auto-populated by sampling collection documents.
- **Operator Mapping:** Context-aware operators (e.g., `array-contains` only for array fields).
- **Query Preview:** A real-time display of the generated Firestore query object.

2. AI Natural Language Interface

> *"The AI does not execute code; it populates the UI."*

Users type requests like *"Show users with age below 50"*. The AI parses the intent and maps it directly into the Query Constructor's visual rows, allowing for human verification before execution.

3. High-Performance Bulk Operations

- **Batched Writes:** Automated chunking of operations into groups of 500 to satisfy Firestore limits.
- **Global Actions:** Support for "Update Field for all filtered results" and "Bulk Delete selected."

Development Roadmap

Phase	Focus	Key Deliverables
P1: Foundation	Connectivity	

Phase	Focus	Key Deliverables
		Electron Shell, Auth Manager, Collection Sidebar.
P2: Query Engine	Constructor UI	JQL-style Row Builder, Data Grid (TanStack/AG Grid).
P3: Intelligence	AI Integration	NLP to JSON Parser, Constructor Auto-population logic.
P4: Mass Actions	Bulk CRUD	Batch Update/Delete logic, Progress Tracking UI.